

TikTok Claims Classification Project

Exploratory Data Analysis (EDA) - Executive Summary

Issue / Problem

The TikTok data team aims to develop a machine learning model to classify user-submitted claims. In this phase of the project, the data must be analyzed, explored, cleaned, and structured before building the model.

Response

At this stage, the TikTok data team performed exploratory data analysis to assess the impact of videos on user engagement. They analyzed key metrics such as view count, like count, and comment count to gain insights into user interactions.

Impact

The exploratory data analysis revealed the need for the future claim classification model to handle null values and address the imbalance in opinion video counts by integrating these factors into the model parameters.

Key Insights

The TikTok data team's exploratory analysis highlighted several key considerations for the classification model, including missing values, the balance between "claims" and "opinions," and the overall distribution of data variables. Two major insights emerged:

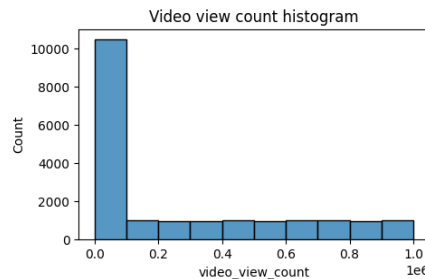
•Null Values

Over 200 null values were identified across seven different columns. To ensure accurate insights, future modeling must account for these missing values rather than assuming a complete dataset. Further investigation is needed to determine the cause of these null values and their potential impact on statistical analysis and model performance.

•Skewed Data Distribution

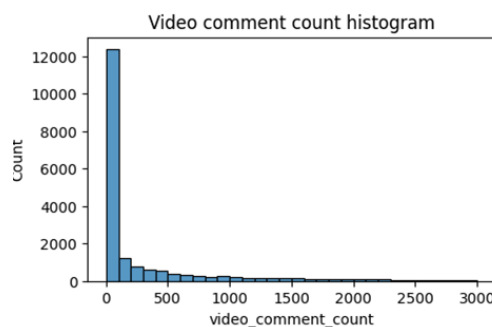
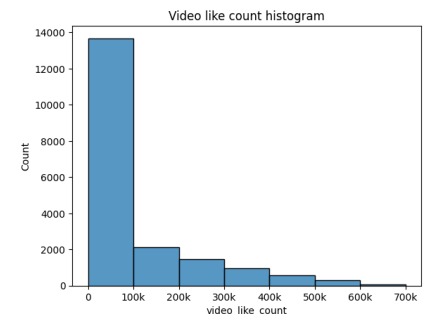
Video views and like counts for opinion videos are heavily concentrated at the lower end, around 1,000, creating a right-skewed distribution. This imbalance will influence the choice of models and modeling techniques used in future analysis.

A crucial aspect of this project's exploratory data analysis is data visualization. The following histograms reveal that most videos are concentrated at the lower end of the value range for three key variables, which reflect TikTok users' engagement



The view count variable is highly skewed, with more than half of the videos receiving fewer than 100,000 views. Beyond this threshold, the distribution becomes more uniform.

Similar to view count, the majority of videos have fewer than 100,000 likes, while those exceeding this threshold are significantly fewer.



Once again, most videos fall at the lower end of the comment count range, with the majority receiving fewer than 100 comments. The distribution is highly right-skewed.